

- Dizziness, syncope, especially when combined with alcohol (contraindicated)



Weight Gain

- Reported but not expected



Sedation

- Many experience and/or can be significant in amount

What to Do About Side Effects

- Wait
- Wait
- Wait
- Review any concomitant medications and consider reducing the dose or discontinuing agents that may be interacting with flibanserin
- Switch to another treatment option

Best Augmenting Agents for Side Effects

- Many side effects cannot be improved with an augmenting agent

Dosing and use

Usual Dosage Range

- 100 mg once daily at bedtime

Dosage Forms

- Tablet 100 mg

How to Dose

- Flibanserin is dosed at bedtime to reduce the risk of hypotension, syncope, and somnolence

Overdose

- Limited experience

Long-Term Use

- Not studied
- However, flibanserin is recommended for long-term use with close monitoring

Habit Forming

- No

How to Stop

- Taper not necessary but may be better tolerated in some patients

Pharmacokinetics

- Metabolized primarily by CYP450 3A4 and to a lesser extent by CYP450 2C19
- Mean terminal half-life approximately 11 hours

Drug Interactions

- The risk of hypotension, syncope, and CNS depression with flibanserin is increased in the presence of alcohol; therefore, the concomitant use of alcohol and flibanserin is contraindicated
- The risk of hypotension, syncope, and CNS depression with flibanserin is increased in the presence of moderate or strong CYP450 3A4 inhibitors; therefore, the concomitant use of moderate/strong CYP450 3A4 inhibitors and flibanserin is contraindicated
- If the patient requires a moderate or strong CYP450 3A4 inhibitor, flibanserin should be discontinued at least 2 days prior to starting the CYP450 3A4 inhibitor
- If the patient is taking a moderate or strong CYP450 3A4 inhibitor, it should be discontinued 2 weeks prior to starting flibanserin
- Concomitant use of flibanserin and weak CYP450 3A4 inhibitors may increase flibanserin exposure, potentially increasing risk of adverse effects
- Concomitant use of flibanserin and CNS depressants may increase the risk of adverse effects such as somnolence

- Concomitant use of flibanserin and CYP450 3A4 inducers may decrease exposure to flibanserin and is not recommended
- Concomitant use of flibanserin and CYP450 2C19 inhibitors may increase flibanserin exposure, potentially increasing risk of adverse effects
- Use of flibanserin in patients who are poor CYP450 2C19 metabolizers may increase flibanserin exposure, potentially increasing risk of adverse effects
- Concomitant use of flibanserin and P-glycoprotein substrates (e.g., digoxin) may increase concentrations of the P-glycoprotein substrate; monitoring is required for P-glycoprotein substrates that have narrow therapeutic index (e.g., digoxin)

Other Warnings/Precautions

- Flibanserin can cause severe hypotension and loss of consciousness, with increased risk in the presence of alcohol or CYP450 3A4 inhibitors; for that reason, flibanserin is available only through a Risk Evaluation and Mitigation Strategy (REMS), which includes elements to assure safe use (ETASU)

Do Not Use

- In the presence of alcohol
- With moderate/strong CYP450 3A4 inhibitors
- In patients with hepatic impairment

- If there is a proven allergy to flibanserin

Special Populations

Renal Impairment

- Exposure may be increased

Hepatic Impairment

- Contraindicated; flibanserin exposure increases 4.5-fold in patients with hepatic impairment

Cardiac Impairment

- Use in patients with cardiac impairment has not been studied

Elderly

- Not approved for use in postmenopausal women
- Some elderly patients may theoretically tolerate lower doses better but this has not been formally studied



Children and Adolescents

- Safety and efficacy have not been established



Pregnancy

- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001
- Controlled studies have not been conducted in pregnant women
- In animal studies, fetal toxicity occurred only in the presence of significant maternal toxicity including reductions in weight gain and sedation
- In animal studies, decreased fetal weight, structural anomalies, and increases in fetal loss occurred with flibanserin at exposures greater than 15 times those achieved with recommended human doses

Breast Feeding

- Unknown if flibanserin is secreted in human breast milk, but all psychotropics are assumed to be secreted in breast milk
- Breast feeding is not recommended during treatment with flibanserin

The art of psychopharmacology

Potential Advantages

- Patients who do not like injections

Potential Disadvantages

- Patients who drink alcohol
- Patients who take concomitant medications that may interact with flibanserin's pharmacokinetic and pharmacodynamic properties

Primary Target Symptoms

- Reduced sexual desire



Pearls

- Flibanserin is not approved for treatment in postmenopausal women or in men and is not indicated to enhance sexual performance
- May act on reward circuits in the CNS to enhance motivation and interest
- If not effective for HSDD, consider avoiding any concomitant use of agents that either enhance serotonin activity (such as SSRIs or SNRIs) or diminish dopamine activity (such as antipsychotics)
- Theoretically, may wish to avoid concomitant use with agents that have potent alpha 1 antagonist actions that may enhance the possibility of hypotension



Suggested Reading

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Stahl SM . Mechanism of action of flibanserin, a multifunctional serotonin agonist and antagonist (MSAA), in hypoactive sexual desire disorder . *CNS Spectr* 2015 ;**20** (1):1 –6 .

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Flumazenil

Therapeutics

Brands

- Romazicon
- Anexate
- Lanexat
- *see index for additional brand names*

Generic?

Yes

Class

- Benzodiazepine receptor antagonist

Commonly Prescribed for

(bold for FDA approved)

- **Reversal of sedative effects of benzodiazepines after general anesthesia has been induced and/or maintained with benzodiazepines**
- **Reversal of sedative effects of benzodiazepines after sedation has been produced with benzodiazepines for diagnostic and**

therapeutic procedures

- **Management of benzodiazepine overdose**
- **Reversal of conscious sedation induced with benzodiazepines (pediatric patients)**

How the Drug Works

- Blocks benzodiazepine receptors at GABA-A ligand-gated chloride channel complex, preventing benzodiazepines from binding there

How Long Until It Works

- Onset of action 1–2 minutes; peak effect 6–10 minutes

If It Works

- * Reverses sedation and psychomotor retardation rapidly, but may not restore memory completely
- * Patients treated for benzodiazepine overdose may experience CNS excitation
- * Patients who receive flumazenil to reverse benzodiazepine effects should be monitored for up to 2 hours for resedation, respiratory depression, or other lingering benzodiazepine effects
- Flumazenil has not been shown to treat hypoventilation due to benzodiazepine treatment

If It Doesn't Work

- Sedation is most likely not due to a benzodiazepine, and treatment with flumazenil should be discontinued and other causes of sedation investigated



Best Augmenting Combos for Partial Response or Treatment Resistance

- None –flumazenil is basically used as a monotherapy antidote to reverse the actions of benzodiazepines

Tests

- None for healthy individuals

Side effects

How Drug Causes Side Effects

- Blocks benzodiazepine receptors at GABA-A ligand-gated chloride channel complex, preventing benzodiazepines from binding there

Notable Side Effects

- May precipitate benzodiazepine withdrawal in patients dependent upon or tolerant to benzodiazepines
- Dizziness, injection site pain, sweating, headache, blurred vision



Life-Threatening or Dangerous Side Effects

- Seizures
- Death (majority occurred in patients with severe underlying disease or who overdosed with non-benzodiazepines)
- Cardiac dysrhythmia

Weight Gain



- Reported but not expected

Sedation



- Reported but not expected
- Patients may experience resedation if the effects of flumazenil wear off before the effects of the benzodiazepine

What to Do About Side Effects

- Monitor patient
- Restrict ambulation because of dizziness, blurred vision, and possibility of resedation

Best Augmenting Agents for Side Effects

- None –augmenting agents are not appropriate to treat side effects associated with flumazenil use

Dosing and use

Usual Dosage Range

- 0.4–1 mg generally causes complete antagonism of therapeutic doses of benzodiazepines
- 1–3 mg generally reverses benzodiazepine overdose

Dosage Forms

- Intravenous 0.1 mg/mL–5 mL multiple-use vial, 10 mL multiple-use vial

How to Dose

- Conscious sedation, general anesthesia: 0.2 mg (2 mL) over 15 seconds; can administer 0.2 mg again after 45 seconds; can administer 0.2 mg each additional 60 seconds; maximum 1 mg
- Benzodiazepine overdose: 0.2 mg over 30 seconds; can administer 0.3 mg over next 30 seconds; can administer 0.5 mg over 30 seconds after 1 minute; maximum 5 mg



Dosing Tips

- May need to administer follow-up doses to reverse actions of benzodiazepines that have a longer half-life than flumazenil (i.e., longer than 1 hour)

Overdose

- Anxiety, agitation, increased muscle tone, hyperesthesia, convulsions

Long-Term Use

- Not a long-term treatment

Habit Forming

- No

How to Stop

- N/A

Pharmacokinetics

- Terminal half-life 41–79 minutes



Drug Interactions

- Food increases its clearance



Other Warnings/Precautions

- Flumazenil may induce seizures, particularly in patients tolerant to or dependent on benzodiazepines, or who have overdosed on cyclic antidepressants, received recent/repeated doses of parenteral benzodiazepines, or have jerking or convulsion during overdose

- Patients dependent on benzodiazepines or receiving benzodiazepines to suppress seizures in cyclic antidepressant overdose should receive the minimally effective dose of flumazenil
- Use with caution in patients with head injury
- Greater risk of resedation if administered to a patient who took a long-acting benzodiazepine or a large dose of a short-acting benzodiazepine
- Flumazenil may induce panic attacks in patients with panic disorder
- Use with caution in cases of mixed overdose because toxic effects of other drugs used in overdose (e.g., convulsions) may appear when the effects of the benzodiazepine are reversed

Do Not Use

- Should not be used until after effects of neuromuscular blockers have been reversed
- If benzodiazepine was prescribed to control a life-threatening condition (e.g., status epilepticus, intracranial pressure)
- If there is a high risk of seizure
- If patient exhibits signs of serious cyclic antidepressant overdose
- If there is a proven allergy to flumazenil or benzodiazepines

Special populations

Renal Impairment

- Dosage adjustment may not be necessary

Hepatic Impairment

- Prolongation of half-life
- Moderate: clearance reduced by half
- Severe: clearance reduced by three-quarters

Cardiac Impairment

- Dosage adjustment may not be necessary

Elderly

- Dosage adjustment may not be necessary



Children and Adolescents

- More variability of pharmacokinetics than in adults
- Safety and efficacy established for reversal of conscious sedation for children over age 1
- Initial 0.01 mg/kg (up to 0.2 mg) over 15 seconds; same dosing pattern as adults; maximum 0.05 mg/kg or 1 mg
- Safety and efficacy for reversal of benzodiazepine overdose, general anesthesia induction or resuscitation of a newborn have not been

established, but anecdotal data suggest similar safety and efficacy as for conscious sedation

Pregnancy

- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001
- Controlled studies have not been conducted in pregnant women
- Not recommended to treat the effects of benzodiazepines during labor and delivery because the effects on the infant have not been studied

Breast Feeding

- Unknown if flumazenil is secreted in human breast milk, but all psychotropics are assumed to be secreted in breast milk
- If treatment with flumazenil is necessary, it should be administered with caution

The art of psychopharmacology

Potential Advantages

- To reverse a low dose of a short-acting benzodiazepine

Potential Disadvantages

- May be too short-acting

Primary Target Symptoms

- Effects of benzodiazepines
- Sedative effects
- Recall and psychomotor impairments
- Ventilatory depression

Pearls

- Can precipitate benzodiazepine withdrawal seizures
- * Can wear off before the benzodiazepine it is reversing
- * Can precipitate anxiety or panic in conscious patients with anxiety disorders

Suggested Reading

Malizia AL , Nutt DJ . The effects of flumazenil in neuropsychiatric disorders . *Clin Neuropharmacol* 1995 ;**18** :215–32 .

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McCloy RF . Reversal of conscious sedation by flumazenil: current status
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Anaesthesiol Scand Suppl* 1995 ;**108** :15 –22 .
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Scand Suppl* 1995 ;**108** :3 –14.

Flunitrazepam

Therapeutics

Brands

- Rohypnol
- *see index for additional brand names*

Generic?

No

Class

- Neuroscience-based Nomenclature: GABA positive allosteric modulator (GABA-PAM)
- Benzodiazepine (hypnotic)

Commonly Prescribed for

(bold for FDA approved)

- Short-term treatment of insomnia (severe, disabling)
- Catatonia

How the Drug Works

- Binds to benzodiazepine receptors at the GABA-A ligand-gated chloride channel complex
- Enhances the inhibitory effects of GABA
- Boosts chloride conductance through GABA-regulated channels
- Inhibitory actions in sleep centers may provide sedative hypnotic effects

How Long Until It Works

- Generally takes effect in less than an hour

If It Works

- Improves quality of sleep
- Effects on total wake-time and number of nighttime awakenings may be decreased over time

If It Doesn't Work

- If insomnia does not improve after 7–10 days, it may be a manifestation of a primary psychiatric or physical illness such as obstructive sleep apnea or restless leg syndrome, which requires independent evaluation
- Increase the dose
- Improve sleep hygiene
- Switch to another agent

Best Augmenting Combos for Partial Response or Treatment Resistance

- Generally, best to switch to another agent
- Trazodone
- Agents with antihistamine actions (e.g., diphenhydramine, TCAs)

Tests

- In patients with seizure disorders, concomitant medical illness, and/or those with multiple concomitant long-term medications, periodic liver tests and blood counts may be prudent

Side Effects

How Drug Causes Side Effects

- Same mechanism for side effects as for therapeutic effects –namely due to excessive actions at benzodiazepine receptors
- Actions at benzodiazepine receptors that carry over to the next day can cause daytime sedation, amnesia, and ataxia
- Long-term adaptations in benzodiazepine receptors may explain the development of dependence, tolerance, and withdrawal

Notable Side Effects

- * Sedation, fatigue, depression
- * Dizziness, ataxia, slurred speech, weakness

- * Forgetfulness, confusion
- * Hyperexcitability, nervousness
- Rare hallucinations, mania
- Rare hypotension
- Hypersalivation, dry mouth
- Rebound insomnia when withdrawing from long-term treatment



Life-Threatening or Dangerous Side Effects

- Respiratory depression, especially when taken with CNS depressants in overdose
- Rare hepatic dysfunction, renal dysfunction, blood dyscrasias

Weight Gain



- Reported but not expected

Sedation



- Many experience and/or can be significant in amount

What to Do About Side Effects

- Wait

- To avoid problems with memory, only take flunitrazepam if planning to have a full night's sleep
- Lower the dose
- Switch to a shorter-acting sedative hypnotic
- Switch to a non-benzodiazepine hypnotic
- Administer flumazenil if side effects are severe or life-threatening

Best Augmenting Agents for Side Effects

- Many side effects cannot be improved with an augmenting agent

Dosing And Use

Usual Dosage Range

- 0.5–1 mg/day at bedtime

Dosage Forms

- Tablet 0.5 mg, 1 mg, 2 mg, 4 mg

How to Dose

- Initial 0.5–1 mg/day at bedtime; maximum generally 2 mg/day at bedtime



Dosing Tips

- Use lowest possible effective dose and assess need for continued treatment regularly
- Flunitrazepam should generally not be prescribed in quantities greater than a 1-month supply
- Patients with lower body weights may require lower doses
- Risk of dependence may increase with dose and duration of treatment
- Use doses over 1 mg only in exceptional circumstances
- Patients who request or who require doses over 1 mg may be more likely to have present or past substance abuse
- Flunitrazepam is 10 times more potent than diazepam

Overdose

- Sedation, slurred speech, poor coordination, confusion, coma, respiratory depression

Long-Term Use

- Not generally intended for long-term use
- Use is not recommended to exceed 4 weeks

Habit Forming

- Some patients may develop dependence and/or tolerance; risk may be greater with higher doses